Exporting Figures in R

You can save plots directly to your computer using functions such as png(), jpeg(), pdf(), and tiff(). These functions open a graphics device (AKA portal) that directs your plots to a file instead of displaying them in the RStudio Plots pane.

Steps:

1. Call a graphics device function (e.g., png()) and specify the file name and any settings.
2. Create your plot using plot(), ggplot(), or another plotting function or supply an object that has your plot stored in it.
3. Call dev.off() to close the graphics device (AKA close the portal) and finish saving the file.

```
library(ggplot2)

# 1. Save the plot as an object
my_plot <- ggplot(iris, aes(x = Sepal.Length, y = Petal.Length)) +
  geom_point() +
  labs(
    title = "Sepal Length vs. Petal Length",
    x = "Sepal Length",
    y = "Petal Length"
  )

# 2. Open a graphics device
png("my_figure.png", width = 800, height = 600)

# 3. Print the plot object to the file
print(my_plot)

# 4. Close the device to save the file
dev.off()
```

Common file formats

Function	File format	Best for
png()	PNG	Digital reports and presentations
jpeg()	JPEG/JPG	Photographs and continuous-tone images
pdf()	PDF	High-quality, scalable figures for papers and reports
tiff()	TIFF	Publications requiring high-resolution images

Important tips

- File location: Files are saved to your current working directory unless you specify a full file path. Use getwd() to check your working directory.
- Specify a path: For example, png("figures/my_figure.png") saves the file in a figures folder inside your working directory. The folder must already exist.
- Always use dev.off(): This closes the graphics device (AKA portal) and ensures the file is finalized properly.
- Set dimensions and resolution: For raster formats like PNG and JPEG, use width, height, and res to control image size and resolution. For example, png("figure.png", width = 6, height = 4, units = "in", res = 300).
- One device (AKA portal), one plot: Each graphics device generally captures the plots drawn while it is open. Close it with dev.off() before opening another device.

Remember: Open the device (AKA portal) → create the plot → close the device (AKA portal).